

Output Devices

Lesson 14 — Displays, Audio and Hard vs Soft Copy

JKSSB Computer Awareness

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What Is an Output Device?

- ▶ An output device **delivers the result** of processing to the user or to another device.
- ▶ It converts machine-readable data into a form a human can **see, hear, or read**.
- ▶ Common examples: monitor, projector, speaker, headphones, printer, and plotter.
- ▶ Output is the last step of the input–process–output cycle.

Definition

An output device receives processed data from the CPU and presents it to the user in a readable form.

Classifying Output Devices

Category	Typical devices
Visual / soft copy	Monitor, projector
Audio	Speaker, headphones
Printed / hard copy	Printer, plotter

A monitor and a projector show output on a screen; a printer and a plotter put it on paper.

Soft Copy vs Hard Copy

Point	Soft copy	Hard copy
Form	Electronic, on a screen	Physical, on paper
Examples	Monitor, projector	Printer, plotter
Editable	Easily edited	Fixed once printed

Trap alert

Soft copy is displayed on a screen and is not printed; **hard copy** is printed on paper. Do not swap the two.

Monitors

- ▶ A **monitor** is the main visual output device of a computer.
- ▶ It displays text, images, and video as **soft copy**.
- ▶ Screen size is measured **diagonally in inches**; picture sharpness is measured in **pixels** (resolution).
- ▶ Common technologies are **CRT**, **LCD**, **LED**, and **OLED**.

Key idea

A monitor shows output to the user; it does not take input (a touchscreen is the exception, because it also senses touch).

CRT vs Flat-Panel Displays

Point	CRT	Flat panel
Full form	Cathode Ray Tube	LCD / LED / OLED
Shape	Bulky, deep	Thin, flat, light
Technology	Electron beam in a vacuum tube	Liquid crystal or self-emitting pixels

Trap alert

A **CRT** (Cathode Ray Tube) is the older, bulky display. Modern monitors are **flat panels**.

LCD, LED, and OLED

Type	What it means	Backlight
LCD	Liquid Crystal Display	Needs a backlight
LED	Light Emitting Diode display (an LCD panel lit by LEDs)	LED backlight
OLED	Organic Light Emitting Diode	Self-emitting, no backlight

In an LED display the LEDs provide the light; in an OLED display each pixel makes its own light.

Projector

- ▶ A **projector** throws a large image onto a screen or wall.
- ▶ It is used in classrooms, meetings, and cinemas.
- ▶ The projected image is **soft copy**, because it is shown, not printed.
- ▶ A projector is an output device; it does not capture anything.

Audio Output: Speakers and Headphones

- ▶ A **speaker** converts an electrical signal into sound for a group of listeners.
- ▶ **Headphones** produce sound for a **single listener** and are worn on or over the ears.
- ▶ Both are **audio output** devices.

Pair to remember

A **microphone** is audio **input**; a speaker or headphone is audio **output**.

Printers: Overview

- ▶ A **printer** produces **hard copy** on paper.
- ▶ Printers fall into two broad classes: **impact** and **non-impact**.
- ▶ Common types: **dot-matrix** (impact); **inkjet** and **laser** (non-impact).
- ▶ Lasers are common in offices; inkjets are common in homes.
- ▶ A printer is an output device (TRAP-12).

Impact vs Non-Impact Printers

Point	Impact	Non-impact
Method	Strikes a ribbon and the paper	No striking; uses ink, toner, or heat
Examples	Dot-matrix	Inkjet, laser
Noise	Noisier	Quieter
Graphics quality	Lower	Higher

Trap alert

A **dot-matrix** printer is an **impact** printer; **inkjet** and **laser** printers are **non-impact**.

Plotter (PYQ-29)

- ▶ A **plotter** draws large, precise **line drawings** such as engineering and architectural designs.
- ▶ It draws using a pen or a vector-based method, on paper larger than a normal printer's.
- ▶ A plotter is an output device, not an input device (TRAP-12).

PYQ-29

Which output device is used to produce **large engineering drawings**? Answer: a **plotter**.

Multifunction Printer (MFP)

- ▶ **MFP** stands for **Multifunction Printer**; it is also called an **all-in-one** device.
- ▶ It combines **printing, scanning, copying**, and often **faxing** in a single unit.
- ▶ Its scanning part is **input**; its printing part is **output**.
- ▶ Common in offices and small businesses.

Keep it straight

An MFP both takes input (when scanning) and gives output (when printing).

Trap Alert: Output or Input? (TRAP-12)

- ▶ **Output**: monitor, projector, speaker, headphones, printer, plotter, drum printer, sound card.
- ▶ **Input**: keyboard, mouse, scanner, light pen, digitizer tablet, punch-card reader, microphone.
- ▶ A plotter and a printer give output; a scanner, light pen, and digitizer give input.

Trap alert

A plotter, a printer, and a sound card are **output**. A scanner and a digitizer are **input**, even when grouped together in one option list.

Near-Neighbours to Keep Apart

Pair	The distinction to state
Printer vs plotter	Printer: standard pages of text/graphics. Plotter: large, precise line drawings.
Monitor vs projector	Monitor: image on its own screen. Projector: image thrown onto a screen or wall.
Speaker vs microphone	Speaker: audio output. Microphone: audio input.
Soft copy vs hard copy	Soft copy: shown on a screen. Hard copy: printed on paper.

Lesson 14: Recall

- ▶ An output device presents processed data in a form the user can see, hear, or read.
- ▶ Soft copy is on a screen (monitor, projector); hard copy is on paper (printer, plotter).
- ▶ Monitor types: CRT (bulky, Cathode Ray Tube); flat panels LCD (backlight), LED (LED backlight), OLED (self-emitting).
- ▶ Audio output: speakers and headphones. A microphone is audio input.
- ▶ Printers: impact (dot-matrix) vs non-impact (inkjet, laser). A plotter draws large engineering drawings (PYQ-29).
- ▶ MFP = Multifunction Printer (print, scan, copy, often fax).
- ▶ TRAP-12: plotter, printer, and sound card are output; scanner, light pen, and digitizer are input.